MATHEMATICS - I

UNIT-I

Summation of series: Binomial series - Exponential series - Logarithmic series - Simple Problems.

RTB(1): Chapter 2: Sections: 2.1, 2.3, 2.4, 2.6.

UNIT-II

Matrices: Symmetric - Skew-Symmetric - Hermitian - Skew - Hermitian - Orthogonal and Unitary matrices - Cayley-Hamilton theorem (without proof) - Verification- Computation of inverse of matrix using Cayley -Hamilton theorem.

RTB(1): Chapter 3: Sections: 3.1, 3.6.

UNIT-III

Numerical Methods: Newton's method to find a root approximately.

Finite Differences: Interpolation: Operators, $\triangle$, V,E, E^{-1} difference tables. Interpolation formulae: Newton's forward and backward interpolation formulae for equal intervals, Lagrange's interpolation formula.

RTB (1): Chapter 4: Section: 4.7.

Chapter 5: Sections: 5.1 and 5.2.

UNIT-IV

Trigonometry: Expansions of sin", cos" in a series ofpowers of sine and cose - Expansions of sin(ne) and cos(ne) in a series sines and cosines of multiples of "0" - Expansions of sine, cose and tane in a series of powers of "0" - Hyperbolic and inverse hyperbolic functions.

RTB (1): Chapter 6: Sections: 6.1 – 6.5

UNIT-V

Differential Calculus: Successive differentiation, nth derivatives,

Leibnitz theorem (without proof) and applications, Jacobians, maxima and minima of functions of two variables- Simple problems

RTB (2): Chapter 1: Sections: 1.1.1, 1.1.2, 1.1.3, 1.1.4, 1.2.1, 1.3.1.